



Risk Management and Process Optimization in Industry 4.0: Integrating Sensors with Critical Path and FMEA

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Abstract. In the era of Industry 4.0, there is a strong focus on automation, advanced smart technologies, and addressing emerging issues related to sustainability and efficient process management. This paper presents a new approach that integrates traditional risk analysis methods, such as Failure Mode and Effects Analysis (FMEA), with the Critical Path Method (CPM) and Business Process Model and Notation (BPMN). The contribution of this method is presented using a selected sample manufacturing process, namely pallet manufacturing. The goal is to show how sensors that monitor machine operation improve process optimisation and manufacturing system sustainability and function.

Utilizing the hybrid FMEA-CPM approach and modelling the production line in BPMN allowed for a comprehensive examination of each activity. This analysis was compared with the outcomes obtained both with and without sensor integration. Sensor data were used for quality control at critical stages of the manufacturing process, reducing the risk of errors in the final product. The results show an over 80% reduction in risk (measured by RPN) and more than 80% improvement in process efficiency, significantly improving decision-making and risk management.

This approach offers valuable insights into how incorporating modern technologies in Industry 4.0 can enhance the administration and control of complex production systems, thus improving efficiency and sustainability.

Keywords: BPMN · Industry 4.0 · Process risk analysis · CPM

1 Introduction

The manufacturing sector today faces increasing demands for efficiency, flexibility, and sustainability. Traditional manufacturing processes are no longer sufficient to maintain competitiveness and respond to the dynamically changing market requirements [5, 6]. Industry 4.0, often referred to as the fourth industrial revolution, is currently the most recognized concept driving the transformation of industries. It is characterized by the integration of advanced digital

technologies such as the Internet of Things (IoT), cloud computing, big data, and smart sensors into manufacturing processes [2, 6, 7]. The advent of Industry 4.0 has brought significant advances in automation, process optimization, and risk management, with an emphasis on sustainability [2, 6, 7]. Within this rapidly evolving landscape, manufacturers are challenged to maintain high production standards while simultaneously reducing risks and inefficiencies. To address these challenges, companies are increasingly choosing the path of innovation. One of the most integrated technologies is sensor-based monitoring systems that collect real-time data on the operation and condition of machines and product quality [3, 9]. These systems, when integrated with well-established risk analysis methods like Failure Mode and Effects Analysis (FMEA) [8] and the Critical Path Method (CPM), offer a powerful framework for optimizing manufacturing processes [4, 8].

This study presents an innovative approach that combines FMEA and CPM with Business Process Model and Notation (BPMN) to comprehensively analyze the production process. By integrating real-time sensor data, the study aims to evaluate the impact of Industry 4.0 technologies on process optimization, risk management, and long-term sustainability. The primary objective of this study is to demonstrate how integrating sensors into the production process can significantly reduce risks, improve process efficiency, and enhance decision-making. The results of this analysis are compared to traditional processes without sensor integration, offering valuable insights into the potential of Industry 4.0 technologies to revolutionize manufacturing and process optimization.

The article examines how Industry 4.0 technologies, particularly sensors, improve manufacturing risk management and process optimisation. Section 1 highlights the need for efficiency and sustainability in manufacturing, while Sect. 2 introduces essential methods such as FMEA, CPM, and BPMN. Real-time sensor data supports risk management and condition-based maintenance with these methods and sensor integration. Sensors were used to monitor machine performance, build and analyse the BPMN model, assign Risk Priority Numbers (RPNs), and optimise critical paths with CPM in Sect. 3. Integration of sensors leads to significant risk reduction and process efficiency improvements, as evidenced by Key Performance Indicators (KPIs) such as RPN and Critical Path Number (CPN). Discussion covers sensor integration's wider implications, Industry 4.0's automation and real-time monitoring potential, and the study's limitations, including cost and applicability to other industries. The study's Conclusion attests to improved decision-making and downtime, while Future Research Directions suggest examining AI-driven predictive maintenance and manufacturing sensor deployment economics.

2 Methodology

The methodology employed in this study integrates several process management and risk assessment techniques to evaluate both current and optimized manufacturing conditions. The hybrid approach combines FMEA to identify

potential points of failure, assign RPNs, and prioritize critical areas within the production process. The CPM is then used to map out the production timeline and highlight the stages where delays or failures would have the greatest impact on the overall process. Additionally, the entire production process is visualized using BPMN, allowing for a comprehensive workflow analysis and identification of areas where automation and smart technologies can improve efficiency and risk management. A key innovation in this methodology is the integration of sensor-based monitoring systems, which provide real-time data on machine performance and product quality. This data leads to improved process management and supports Condition-Based Maintenance (CBM) strategies, reducing machine downtime and improving long-term sustainability [1, 12]. The methodology is designed to compare outcomes from traditional processes without sensor integration against those using real-time sensor data, providing insights into the benefits of Industry 4.0 technologies in risk management and process optimization.

2.1 Failure Mode and Effects Analysis (FMEA)

FMEA is a systematic method for identifying potential failures in a process and assessing their impact on the system. It helps prioritize risks based on their severity, frequency, and detectability. In our study, FMEA was utilized to identify and prioritize potential points of failure in the pallet manufacturing process. Each step, from sorting raw wood to final quality inspections, was evaluated for possible failures. Severity, occurrence, and detection ratings were assigned to each activity, allowing for the calculation of RPNs. This method helps identify the most vulnerable stages of the process, offering a framework to mitigate risks and inefficiencies.

For instance, during the cutting stage, a potential failure mode was identified as inaccurate cuts due to blade wear. The severity (S) was rated as 7 (on a scale of 1–10) because improper cuts can lead to structural weaknesses in pallets. The occurrence (O) was rated as 5 due to the moderate likelihood of blade wear over time. The detection (D) was rated as 4 since blade wear can be detected through regular inspections. This resulted in an RPN.

FMEA is being utilised by researchers mostly in order to analyse manufacturing and assembly processes, with the goal of identifying critical parameters. Wu et al. [13] emphasise in their paper that when FMEA is applied to advanced manufacturing, in addition to ensuring the completeness of recognition when identifying failure modes, it is necessary to eliminate “boring and complicated human activities” and implement the automatic recognition process as soon as possible to improve FMEA application efficiency.

One limitation encountered was the subjectivity in assigning S, O and D values, which we mitigated by involving a cross-functional team to reach a consensus. Additionally, detecting certain subtle failure modes required more advanced monitoring than initially available.

2.2 Critical Path Method (CPM)

CPM is a step-by-step project management technique for process planning that defines both critical and non-critical tasks to prevent time-frame issues and process bottlenecks.

The CPM was used to map out the production process, highlighting the most critical stages where delays or failures would have the greatest impact on the overall production time. By combining CPM with FMEA, the study identified critical paths and evaluated how risks at various stages could affect the overall production timeline. Time estimates for each activity were provided, allowing for more accurate workflow optimization.

2.3 Business Process Model and Notation (BPMN)

BPMN is a graphical representation for specifying business processes in a workflow. The entire production process was visualised using BPMN. This allowed an in-depth workflow analysis to identify areas where automation and smart technologies could improve efficiency and risk management. In our study, BPMN was utilized to create a detailed visual model of the pallet manufacturing process.

One benefit of using BPMN was the ability to communicate complex process flows clearly to stakeholders from different backgrounds. However, a challenge encountered was ensuring that the BPMN diagram remained clear and uncluttered despite the complexity of the process.

We can be inspired by Cardoso et al. (2021) [4], which added quantitative risk assessment to BPMN. BPMN models enhance operational efficiency, adaptability, and competitiveness. Participants and stakeholders in the business process can easily understand the visual representation of BPMN steps. Furthermore, it highlights the individuals engaged in the process of optimisation [10, 11]. Following this, the BPMN model was used to compare the process with and without sensor integration, showing how technological upgrades affect workflow.

2.4 Sensor Integration and Data Collection

To enhance the manufacturing process, sensor-based monitoring systems were introduced at various stages of production. These sensors captured real-time data on machine performance, product quality, and operating conditions. Various sensors were integrated at critical stages. For instance:

- **Optical Scanners:** Used during the sorting stage to assess wood quality by detecting surface defects.
- **Infrared Sensors:** Employed to measure moisture content in the wood, crucial for preventing warping.
- **Force Sensors:** Installed in assembly machines to ensure nails and screws are applied with correct pressure.

One challenge was managing the large volume of data generated, which required robust data processing capabilities. Additionally, integrating sensors with legacy equipment required custom interfaces.

3 Implementation

In this section, the implementation of the process analysis, improvement, and evaluation is meticulously examined through the use of numerous structured algorithms. This section is designed to outline the procedures for the development, analysis, enhancement, and reassessment of a BPMN process model through the application of a variety of methodologies, such as FMEA, Min-Max normalisation, CPM, and sensor implementation.

In order to accomplish the overarching objective of process optimisation, the entire implementation is partitioned into numerous critical stages, each of which is represented by a distinct algorithm that executes specific tasks. Detailed descriptions of each algorithm are provided in the subsequent sections, which illustrate the algorithm's inputs, operations, and outputs.

3.1 Model Preparation

In this stage, the BPMN process model is built and initialized. The algorithm focuses on defining the activities within the process, along with their relationships, such as predecessors and successors. This model serves as the foundation for subsequent analysis and evaluation. The detailed steps are described in the algorithm 1.

The BPMN model was prepared using the bpmn.io, built and maintained by Camunda and contributors, which allowed for the intuitive creation and editing of process diagrams. We began by listing all the activities involved in the pallet manufacturing process, such as “Receiving Raw Wood” and “Final Quality Inspection.”

Each activity was assigned attributes including:

- **Duration:** Estimated time to complete the activity.
- **Resources Required:** Machines, labor, and materials.
- **Predecessors and Successors:** Defining the flow and dependencies.

For example, “Cutting to Required Dimensions” cannot begin until “Sorting by Quality” is completed. This hierarchical structuring facilitated accurate mapping of the process flow.

One challenge was ensuring that all possible paths, including rework loops, were accurately represented, which required iterative refinement of the model.

3.2 Model Analysis

The BPMN model is subjected to a comprehensive analysis following its preparation. This phase entails the assignment of RPN to each activity in accordance with the FMEA methodology. Furthermore, the duration of each activity is ascertained, and the Min-Max normalisation method is employed to scale both the time and RPN values to a shared range. This analysis is essential for comprehending the risks and duration of the process activities.

For instance, the activity “Assembling Pallets” had an original RPN of 165 and a duration of 10 min. After scaling:

Algorithm 1. Process Analysis with Sensor Implementation using BPMN, FMEA, and CPM

```

1: Creation of BPMN Model
2: procedure CREATEBPMNMODEL
3:   initialize process
4:   for each activity in the process do
5:     define activity (name, order, predecessors/successors)
6:   end for
7:   return BPMN_model
8: end procedure

```

- **Scaled RPN:** scaled_RPN = 8.5
- **Scaled Time:** scaled_time = 9.0

This indicated that “Assembling Pallets” was both high-risk and time-consuming, making it a prime candidate for improvement.

The analysis revealed that activities with high RPNs were often on the critical path, emphasizing the need for targeted interventions in these areas.

The analysis steps are detailed in the algorithm 2.

3.3 Model Improvement

In this stage, the initial analysis results are used to identify critical activities that significantly affect the overall process using the CPM. Based on these results, sensors are implemented in the most critical parts of the process to monitor performance and manage risks. This improvement phase is described in the algorithm 3.

3.4 Improved Model Analysis

Once sensors are implemented, the process model is analyzed again to evaluate the impact of the improvements. The process repeats the FMEA analysis, Min-Max scaling, and activity value calculation. The results of this analysis help in understanding the effectiveness of sensor implementation. The steps of this reanalysis are detailed in the algorithm 4.

3.5 Final Evaluation

The final step in the implementation is to compare the results of the initial process model with the improved model that includes sensor implementation. This comparison focuses on changes in risk (RPN values), time, and critical paths, and it evaluates whether the sensor implementation has brought the desired improvements. The final comparison is described in the algorithm 5.

Algorithm 2. Analysis of the BPMN Model using FMEA and Min-Max Scaling

```

1: Assign RPN
2: procedure ASSIGNRPN(BPMN_model)
3:   for each activity in BPMN_model do
4:     get severity ( $S$ ), occurrence ( $O$ ), and detection ( $D$ ) values
5:     calculate
                                      $RPN = S \times O \times D$  (1)
6:     assign  $RPN$  to activity
7:   end for
8: end procedure

9: Assign Duration to Activities
10: procedure ASSIGNEDURATIONS(BPMN_model)
11:   for each activity in BPMN_model do
12:     get duration (time) for each activity
13:     assign duration to activity
14:   end for
15: end procedure

16: Scaling RPN and Time Values to the Same Scale
17: procedure SCALEVALUES(BPMN_model, a, b)
18:   get  $min\_RPN$ ,  $max\_RPN$  from BPMN_model
19:   get  $min\_time$ ,  $max\_time$  from BPMN_model
20:   for each activity in BPMN_model do
21:     // Min-max scaling for time:
22:
                                      $scaled\_time = a + \frac{(time - min\_time)}{(max\_time - min\_time)} \times (b - a)$  (2)
23:     // Min-max scaling for RPN:
24:
                                      $scaled\_RPN = a + \frac{(RPN - min\_RPN)}{(max\_RPN - min\_RPN)} \times (b - a)$  (3)
25:     assign  $scaled\_time$  and  $scaled\_RPN$  to activity
26:   end for
27: end procedure

28: Assign Weights to Each Value
29: procedure ASSIGNWEIGHTS(weight_time, weight_RPN)
30:   return [weight_time, weight_RPN]
31: end procedure

```

3.6 Main Function

The primary function of this algorithm is to establish a connection between the various components of the BPMN model analysis and improvement process. It automates the process of constructing the model, analysing risks and time constraints, implementing sensors, and conducting a subsequent evaluation. This

Algorithm 3. Model Improvement: Sensor Implementation and CPM

```

1: Calculate Total Value of Activity
2: procedure CALCULATETOTALVALUE(BPMN_model, weight_time, weight_RPN)
3:   for each activity in BPMN_model do
4:     total_value = (weight_time  $\times$  scaledTime) + (weight_RPN  $\times$  scaledRPN)
5:     assign total value to activity
6:   end for
7: end procedure

8: Implement Results into CPM
9: procedure IMPLEMENTCPM(BPMN_model)
10:   calculate the critical path based on activity durations and dependencies
11:   mark critical activities
12:   return CPM_model
13: end procedure

14: Implement Sensors Based on Initial Phase Results
15: procedure IMPLEMENTSENSORS(CPM_model)
16:   for each activity in CPM_model do
17:     if RPN or time on the critical path exceeds a threshold then
18:       implement sensor into activity
19:     end if
20:   end for
21: end procedure

```

Algorithm 4. Analysis of the Improved Model

```

1: Repeat Steps 1-7 for Model with Sensors
2: procedure REPEATWITHSENSORS(BPMN_model)
3:   BPMN_model_with_sensors = ImplementSensors(BPMN_model)
4:   repeat AssignRPN_FMEA(BPMN_model_with_sensors)
5:   repeat AssignDurations(BPMN_model_with_sensors)
6:   repeat ScaleValues(BPMN_model_with_sensors)
7:   repeat CalculateTotalValue(BPMN_model_with_sensors, weight_time,
   weight_RPN)
8:   ImplementCPM(BPMN_model_with_sensors)
9: end procedure

```

method enables the optimisation of processes, the mitigation of risks, and the enhancement of system performance.

3.7 Current State

The process of making wooden pallets (see Fig. 1) begins with the receipt of raw wood, which is the primary material used in production. Upon receipt, the wood is first sorted by quality to ensure that only suitable material advances to the next stages of production. After sorting, the wood is cut to the dimensions required for pallet assembly.

Algorithm 5. Final Evaluation of the Sensor Implementation

```

1: Compare Efficiency of Sensor Implementation
2: procedure COMPAREEFFICIENCY(BPMN_model, BPMN_model_with_sensors)
3:   evaluate changes in RPN, time, and total activity value
4:   compare critical paths
5:   if improvement in the model with sensors then
6:     print "Sensor implementation is effective"
7:   else
8:     print "Sensor implementation did not provide the expected improvement"
9:   end if
10: end procedure

```

Algorithm 6. Main Function: Process Analysis and Improvement

```

1: procedure MAIN
2:   BPMN_model = CreateBPMNModel()
3:   AssignRPN_FMEA(BPMN_model)
4:   AssignDurations(BPMN_model)
5:   ScaleValues(BPMN_model, 1, 10) // Example scale [1, 10]
6:   [weight_time, weight_RPN] = AssignWeights(0.5, 0.5) // Example weights
7:   CalculateTotalValue(BPMN_model, weight_time, weight_RPN)
8:   CPM_model = ImplementCPM(BPMN_model)
9:   BPMN_model_with_sensors = ImplementSensors(CPM_model)
10:  RepeatWithSensors(BPMN_model_with_sensors)
11:  CompareEfficiency(BPMN_model, BPMN_model_with_sensors)
12: end procedure

```

Following the cutting process, the wood goes through the first quality control checkpoint, where it is inspected to ensure that it meets the specifications. If the wood is properly cut, it moves on to the assembly stage. If not, it is routed to a defect elimination process, where any identified flaws are corrected. Once the defects have been removed, the wood will proceed to the pallet assembly stage.

During the assembly phase, the wood is formed into pallets. After assembly, the pallets are subjected to a second quality control inspection to ensure that they have been correctly assembled and meet the required standards. If the pallets pass this inspection, they will proceed to the finishing modifications stage. If any problems are discovered, the pallets are sent back for rework to correct any errors in the assembly.

Following any necessary modifications, the pallets are subjected to a final quality control inspection. Before being approved for use, the finished pallets undergo a final inspection to ensure that they meet all specifications. If the pallets pass this final inspection, they will be ready for the next step in the supply chain. However, if they fail, an evaluation is performed to determine whether they can be repaired. If repairable, the pallets are returned to the rework stage. If repair is not possible, the pallets are discarded or repurposed using a liquidation process.

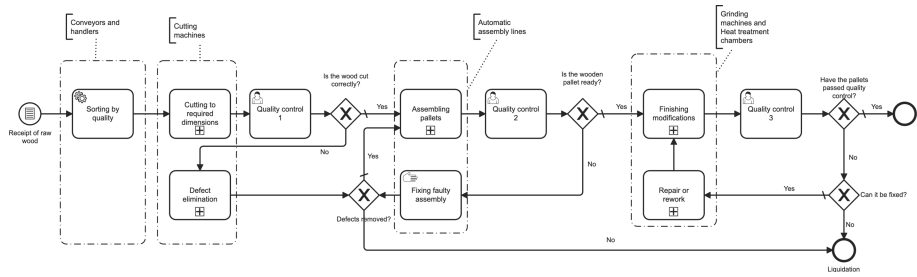


Fig. 1. Process model of the current state (without sensor integration)

The Table 1 that accompanies the process model contains important data for performing FMEA. The table is organised to reflect the likelihood of failures (O - Occurrence), their severity (S), and the ability to detect (D) them before they reach the customer. It also includes an estimated time for completing each activity in the pallet production process.

Each stage of the process—from sorting raw wood by quality, cutting it to the required dimensions, and assembling the pallets to the various quality control checks—has been assessed for risk factors. The FMEA parameters assist in determining which stages of the process are most vulnerable to failure, where such failures could have the greatest impact, and how likely it is that these failures will be detected before causing problems later on.

The time estimates provide a clear picture of how long each step in the process takes, allowing for more effective workflow planning and optimisation. Understanding the risk and time involved in each activity allows for more effective process management, ensuring high-quality output while minimising delays and inefficiencies.

Table 1. Data used in FMEA and estimated activity time without sensor integration

Activity	O	S	D	Min/batch
Sorting by quality	3.5	4.5	2	5
Cutting to required dimensions	4.5	5.5	2	3
Quality control 1	2	9	5.5	5
Assembling pallets	5.5	5.5	5.5	10
Quality control 2	5.5	5.5	4.5	5
Finishing modifications	2	2	2	4
Quality control 3	2	9	9	5

3.8 Industry 4.0 Support Situation

The updated process model(see Fig. 2) makes use of various sensors throughout the wooden pallet manufacturing process to improve quality control and efficiency. The process starts with the receipt of raw wood, which is then sorted by quality. At this point, optical scanners and infrared sensors can be used to assess the quality of the wood by detecting defects such as knots, cracks, or moisture content, ensuring that only high-quality wood moves forward.

The wood is then cut to the required dimensions. Laser measurement sensors and position sensors are used here to ensure cutting precision while reducing waste and the need for rework. Following cutting, the first quality control check-point employs vision systems outfitted with high-resolution cameras to inspect the wood for any inaccuracies or flaws automatically.

If the wood is properly cut, it moves on to the assembly stage; otherwise, it is directed to the defect elimination process. Ultrasonic sensors are used at this stage to detect and repair any internal defects that are not visible from the outside. Once the defects have been removed, the wood will proceed to the pallet assembly phase.

During assembly, force and position sensors integrated into automatic assembly lines ensure that each component is properly fitted with the appropriate force, ensuring consistency and reducing errors. After assembly, the pallets go through a second quality control check, during which 3D scanners or machine vision systems ensure that the pallets are correctly assembled and meet the required standards.

If the pallets pass this inspection, they proceed to the finishing modifications stage. If not, they are returned for rework to correct any assembly flaws, possibly using real-time feedback sensors that direct operators or machines to make precise adjustments.

During the finishing modifications stage, thickness and surface roughness sensors are used to ensure that the pallets meet final specifications such as smoothness and thickness. If problems persist, the pallets are moved to the repair or rework stage, where previously used sensors help ensure that the modifications are up to standard.

Finally, in the third stage of quality control, a comprehensive sensor system is used, which includes load sensors to test strength and durability, as well as thermal imaging cameras to detect any weaknesses. Pallets that pass this final inspection are approved for the next step in the supply chain. If a pallet fails, it is evaluated to see if it can be repaired; if not, it is liquidated.

The Table 2 displays data for the FMEA as well as time estimates for each activity in the wooden pallet manufacturing process, taking into account the changes brought about by sensor integration. The table is set up similarly to the previous one, with columns for Occurrence (O), Severity (S), Detection (D), and the time required for each activity in minutes.

The FMEA values have changed noticeably since the previous table. The sensors' improved precision and real-time monitoring have reduced the likelihood of failure across a wide range of activities. For example, in the cutting and

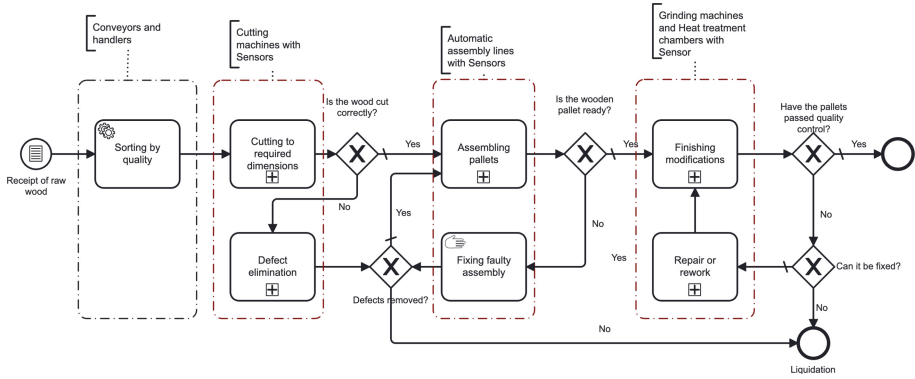


Fig. 2. Process model with sensor integration

assembly stages, the risk of errors has been reduced, resulting in lower occurrence ratings.

However, the severity ratings have risen in some cases, particularly during the cutting and assembly stages. This is due to the fact that, while the sensors improve accuracy, any failure that occurs may have more serious consequences, given the increased reliance on automatic processes. As a result, the impact of potential failures is greater, leading to an increase in severity scores.

The values of detectability remain relatively low, indicating that the sensors effectively detect potential problems before they reach the production line. This ability to detect problems early is critical in preventing defects from reaching the finished product.

The time estimates for each activity vary as well. The time required to sort by quality has decreased, due to efficiency gains from automated quality assessment using sensors. Meanwhile, despite their increased complexity, the cutting and assembly processes maintain a consistent time allocation, aided by the streamlined operations that sensors allow.

Table 2. Data used in FMEA and estimated activity time with sensor integration

Activity	O	S	D	Min/batch
Sorting by quality	3.5	4.5	2	2
Cutting to required dimensions	2.5	7.5	2	3
Assembling pallets	2.5	7.5	2	5
Finishing modifications	2	2	2	2

The Fig. 3 compares two scenarios: one without automation (solid line) and one with sensors integrated into the process (dashed line). The results are assessed using a weighted combination of the RPN, equation 1, from the FMEA

values and time efficiency. The weight distribution varies from 0.1 for RPN and 0.9 for time in Case 1 to 0.9 for RPN and 0.1 for time in the final case.

As the weight for RPN increases, the overall score for the scenario without automation (solid line) rises more dramatically than the scenario with sensors (dashed line). This suggests that, without automation, the process is more heavily influenced by the risk of failure as RPN is prioritised over time. The steep incline of the solid line indicates that as more emphasis is placed on risk management, the deficiencies of the non-automated process become more pronounced, most likely due to increased occurrences and severity of potential failures, as well as lower detection capabilities.

In contrast, the scenario with sensors shows a more gradual increase in score, indicating better risk management performance even as RPN weight increases. Sensor integration is likely to reduce the frequency and severity of potential failures while also improving the ability to detect issues early. This leads to more consistent performance across different weight distributions of RPN and time.

The primary reason for these differences is that automation, via sensor integration, improves the process's ability to effectively manage risks. The sensors provide real-time monitoring and precise adjustments, which significantly reduces the likelihood and severity of failures. As a result, when more weight is given to RPN, the automated process does not show a significant increase in score because the risks are already well managed. In contrast, the non-automated process struggles with risk management, resulting in a steeper rise in score when RPN is prioritised.

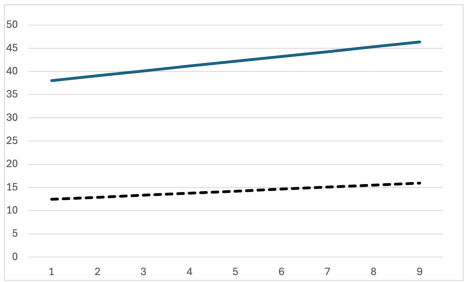


Fig. 3. Comparison of the two situations

3.9 Detailed Evaluation of Key Performance Indicators (KPIs)

The implementation of automation and sensors has resulted in substantial enhancements in process efficiency and risk management, as indicated by the CPN and RPN. The total RPN was 652.5 (normalised to 31.49) prior to the implementation of the sensors. RPN decreased by more than 82% to 114.5 (normalised to 12.83) following the introduction of the sensors. The probability of

serious production process failures is significantly reduced by this decrease, which suggests a fundamental improvement in the area of risk identification and management.

The CPN value was 221.65 (normalised to 40.11) prior to the implementation of the sensors, but it decreased to 42.75 (normalised to 13.32) after the implementation of the sensors, with respect to the overall time efficiency of the process. This decrease in CPN of over 80% is indicative of significant enhancements in process efficiency, as it mitigates the occurrence of delays or errors in the production chain and expedites critical tasks.

4 Results

The results of this study demonstrate significant improvements in both risk management and process efficiency following the integration of sensors into the pallet manufacturing process. These results are presented through a detailed comparison of KPIs, such as RPN and CPN, measured before and after sensor integration.

4.1 Risk Reduction

Before the implementation of sensors, the total RPN for the manufacturing process was 652.5, normalized to 31.49. After integrating sensors, the RPN dropped to 114.5, normalized to 12.83. This represents an 82% reduction in overall risk, indicating that the integration of real-time monitoring through sensors significantly decreased the likelihood of critical failures in the production process.

The largest reductions in RPN were observed in stages related to machine performance monitoring, where sensors provided early detection of potential issues, allowing for immediate corrective actions before any significant disruptions occurred.

4.2 Process Efficiency

Similarly, process efficiency, as measured by the CPN, improved dramatically. Before sensor integration, the total CPN was 221.65, normalized to 40.11. After the implementation of sensors, the CPN dropped to 42.75, normalized to 13.32, representing a more than 80% improvement in efficiency.

This improvement reflects the impact of real-time data collection on the operation and condition of machines, products quality which enabled real-time adjustments and possibly CBM, leading to reduced production delays and minimized downtime.

4.3 Sensor Impact on Critical Path

The integration of sensors had the most pronounced effect on activities located on the critical path, where delays would have the greatest impact on the overall production time. By continuously monitoring these critical activities, sensors

provided real-time data that allowed for immediate corrective measures, preventing delays and ensuring smoother operation. Some of the data could be possibly used for CBM, allowing the system to anticipate and address equipment issues before they resulted in production stoppages.

4.4 Quality Control Improvements

Another key area of improvement was in quality control. The sensors implemented in stages related to product inspection, such as optical scanners and infrared sensors, allowed for more precise detection of defects in raw materials and finished products. This led to a noticeable reduction in the number of defective products, enhancing the overall quality of the output. The real-time data collected by these sensors enabled quick identification of issues, reducing the need for rework and improving the consistency of production.

4.5 Comparative Analysis

Figure 3 compares the overall process performance before and after sensor integration, using both RPN and CPN as metrics. The solid line represents the scenario without sensor integration, where risk management was more dependent on manual processes, and inefficiencies were higher. The dashed line shows the post-sensor integration scenario, where automation and real-time monitoring reduced both the occurrence and severity of potential failures while also improving time efficiency.

5 Conclusion

This study demonstrates the impact that Industry 4.0 technologies, particularly sensor integration, can have on optimizing manufacturing processes and enhancing risk management. By combining traditional methods like FMEA and the CPM with real-time sensor data, the production process can be significantly improved in terms of both efficiency and reliability.

The integration of sensors into critical stages of the pallet manufacturing process resulted in an 82% reduction in RPN, which indicates a substantial decrease in the likelihood of failures and defects. Furthermore, the use of sensors led to more than an 80% improvement in process efficiency, as measured by the CPN. These improvements were driven by the ability of sensors to provide real-time monitoring, which enabled proactive decision-making and CBM.

In addition to improving efficiency and reducing risk, sensor integration enhanced quality control, allowing for more precise detection of defects and improving the consistency of product quality. The use of advanced monitoring technologies enabled faster identification and correction of issues, reducing the need for rework and increasing overall productivity.

This research underscores the potential of Industry 4.0 technologies to revolutionize traditional manufacturing processes by improving sustainability, minimizing risks, and increasing overall efficiency. The findings provide valuable

insights for manufacturers looking to adopt similar technologies to enhance their operations.

5.1 Key Findings

The integration of sensors into the pallet manufacturing process demonstrated significant improvements in both risk management and process efficiency. By combining traditional methodologies such as FMEA and CPM with real-time sensor data, the manufacturing process experienced an 82% reduction in RPNs and an 80% improvement in process efficiency as measured by CPN. These improvements were largely driven by real-time monitoring which allowed for early detection of potential issues and proactive decision-making.

5.2 Contributions to Industry 4.0

This study highlights how the adoption of Industry 4.0 technologies, specifically sensor integration, can lead to revolutionary changes in traditional manufacturing processes. The real-time data collected by sensors not only enhances overall operational control and product quality but also supports CBM. These technologies offer a practical pathway for manufacturers to improve process sustainability, reduce risks, and increase competitiveness in an increasingly digitized industrial landscape.

5.3 Limitations

While the study provides promising results, there are certain limitations to consider.

- The specificity of the pallet manufacturing process may limit the generalizability of the findings. This study focused on a specific production line, which may not fully represent the requirements of other industries or more complex manufacturing systems.
- Production environments often include devices from different vendors that may use different communication protocols. Ensuring compatibility and the ability to communicate between different sensors and machine systems is essential.
- Legacy manufacturing equipment may not be ready for direct integration of modern sensors, which may require additional modifications.
- The initial cost of sensors and infrastructure to process data in real-time may pose a barrier for smaller manufacturers.
- Data security is a concern, as the integration of sensors and IoT devices increases vulnerability to cyber-attacks. Implementing robust cybersecurity measures is essential to protect sensitive operational data.
- Maintenance of sensor equipment presents ongoing costs and operational considerations. Regular calibration and updates are necessary to ensure accuracy, potentially causing minor disruptions.

5.4 Future Research Directions

There is significant potential for future research in optimizing the use of sensors and expanding their application in various industrial settings. Future work could explore the integration of artificial intelligence (AI) for predictive maintenance and process optimization, further enhancing the ability to anticipate and mitigate risks. Additionally, investigating the economic impacts of sensor integration, such as cost-benefit analyses and return on investment (ROI), will be crucial for promoting the widespread adoption of these technologies. Exploring these applications across different industries could also offer valuable insights into the scalability and adaptability of Industry 4.0 technologies.

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